

Data Analytics for Bouldering

Ethan Baron

Intro

Thanks David

About me: UofT (UTSPAN, Zelus) → NYU PhD

This is work with Victor and Zeke which won student repro research competition at CMSAC 2024

Learn about audience: anyone from Thornhill? how many people boulder? How many watch bouldering? Do they know logistic regression?

Collaborators



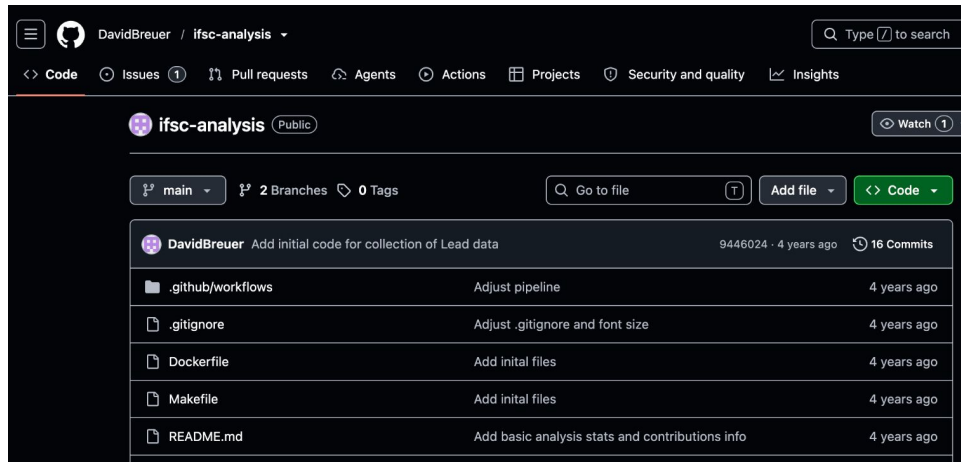
Victor Hau



Zeke Weng

Data

- IFSC
- Years 2007-2022
- 2517 boulder problems



Year	Competition	Problem	Climber	Top?
2018	Munich	Semifinal 1	Adam Ondra	Yes
2020	Paris	Final 4	Janja Garnbret	Yes
2021	Toronto	Qualifying 3	Ethan Baron	No

Who is the best climber?

Top 10 male climbers			
#	Climber	Success %	Count
1	Adam Ondra	85.4%	(532/623)
2	Tatsumi Nitta	85.0%	(17/20)
3	Kilian Fischhuber	84.9%	(956/1126)
4	David Lama	82.8%	(130/157)
5	Dmitrii Sharafutdinov	81.9%	(1007/1229)
6	Cédric Lachat	81.8%	(278/340)
7	Jérôme Meyer	81.6%	(31/38)
8	Paul Robinson	80.9%	(72/89)
9	Tomoa Narasaki	80.6%	(794/985)
10	Kokoro Fujii	80.1%	(910/1136)

Need to account for boulder difficulty!

Bradley-Terry Model

Probability of success depends on climber strength and problem difficulty

$$\Pr(i > j) = \frac{e^{\beta_i}}{e^{\beta_i} + e^{\beta_j}}$$

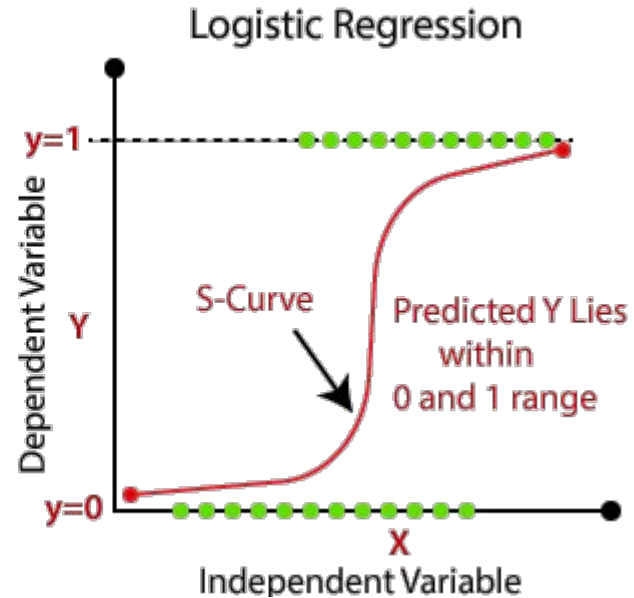
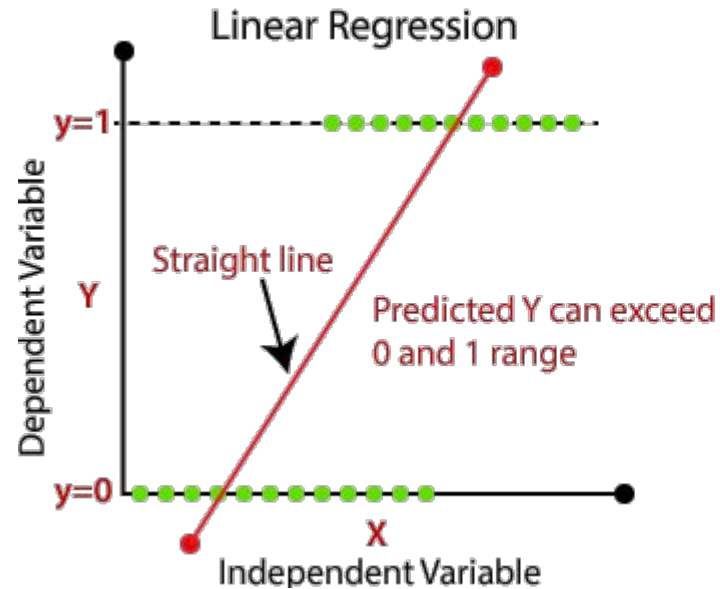
i = climber

j = boulder problem

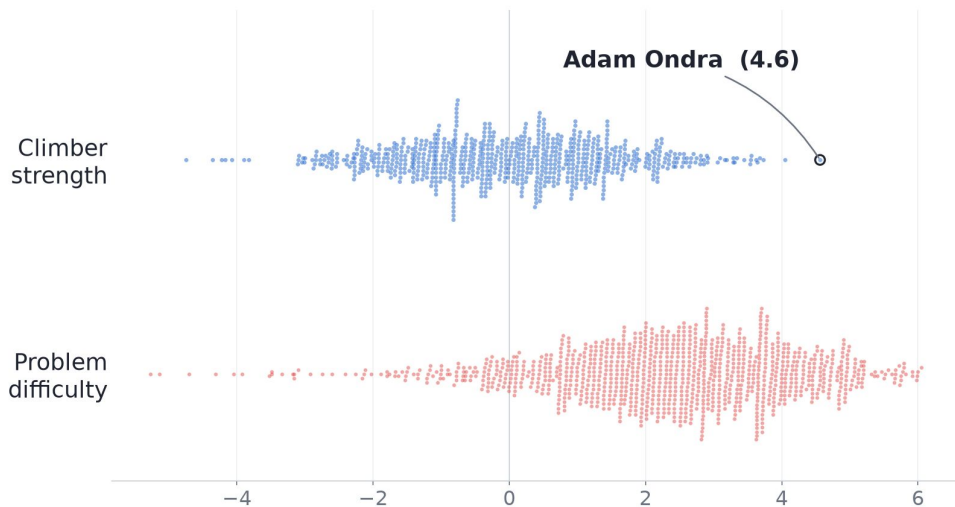
How to get the strengths and difficulties?

It's actually a kind of logistic regression!

Logistic regression



Bradley-Terry Results



#	Climber	Rating	Top %	Count
1	Adam Ondra	4.56	80.0%	244/305
2	Kilian Fischhuber	4.05	78.6%	430/547
3	Jérôme Meyer	3.73	66.7%	12/18
4	David Lama	3.68	71.1%	54/76
5	Tomoa Narasaki	3.64	70.2%	340/484

Still need to
account for
sample size!

Regularization

- Regularization: add “penalty” that pulls ratings towards 0
 - Need lots of data to merit getting big rating!

#	Climber	Rating	Top %	Count
1	Kilian Fischhuber	0.894	78.6%	430/547
2	Dmitrii Sharafutdinov	0.816	73.2%	440/601
3	Rustam Gelmanov	0.782	70.3%	466/663
4	Kokoro Fujii	0.725	70.0%	390/557
5	Tomoa Narasaki	0.683	70.2%	340/484
6	Adam Ondra	0.680	80.0%	244/305
7	Jongwon Chon	0.662	69.4%	331/477
8	Sean McColl	0.624	63.6%	412/648
9	Jakob Schubert	0.622	67.9%	318/468
10	Aleksei Rubtsov	0.616	65.6%	349/532

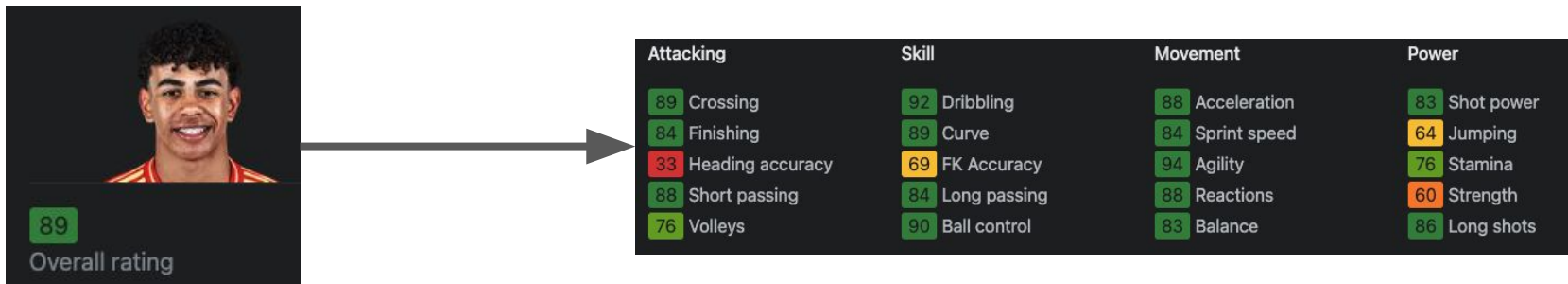
Now top-rated climbers all have
big sample size!

What makes a climber good?



What makes a climber good?

- Different boulder problems require different techniques
 - Different climbers can have different strengths
 - How can we quantify this?
-
- Need multivariate representation!



Vector embeddings

But where do we get these
vector embeddings from?



"apple"
(0.1, 0.5, 0.2)



"cat"
(-0.5, -0.2, 0.4)



"banana"
(0.2, 0.4, 0.2)

New model



Adam Ondra

1.10	2.94	0.04	0.43	1.72
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•

-1.29	-0.36	1.14	-0.44	0.79
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=

-1.3



Ethan Baron

0.49	-2.94	-0.14	3.12	-4.48
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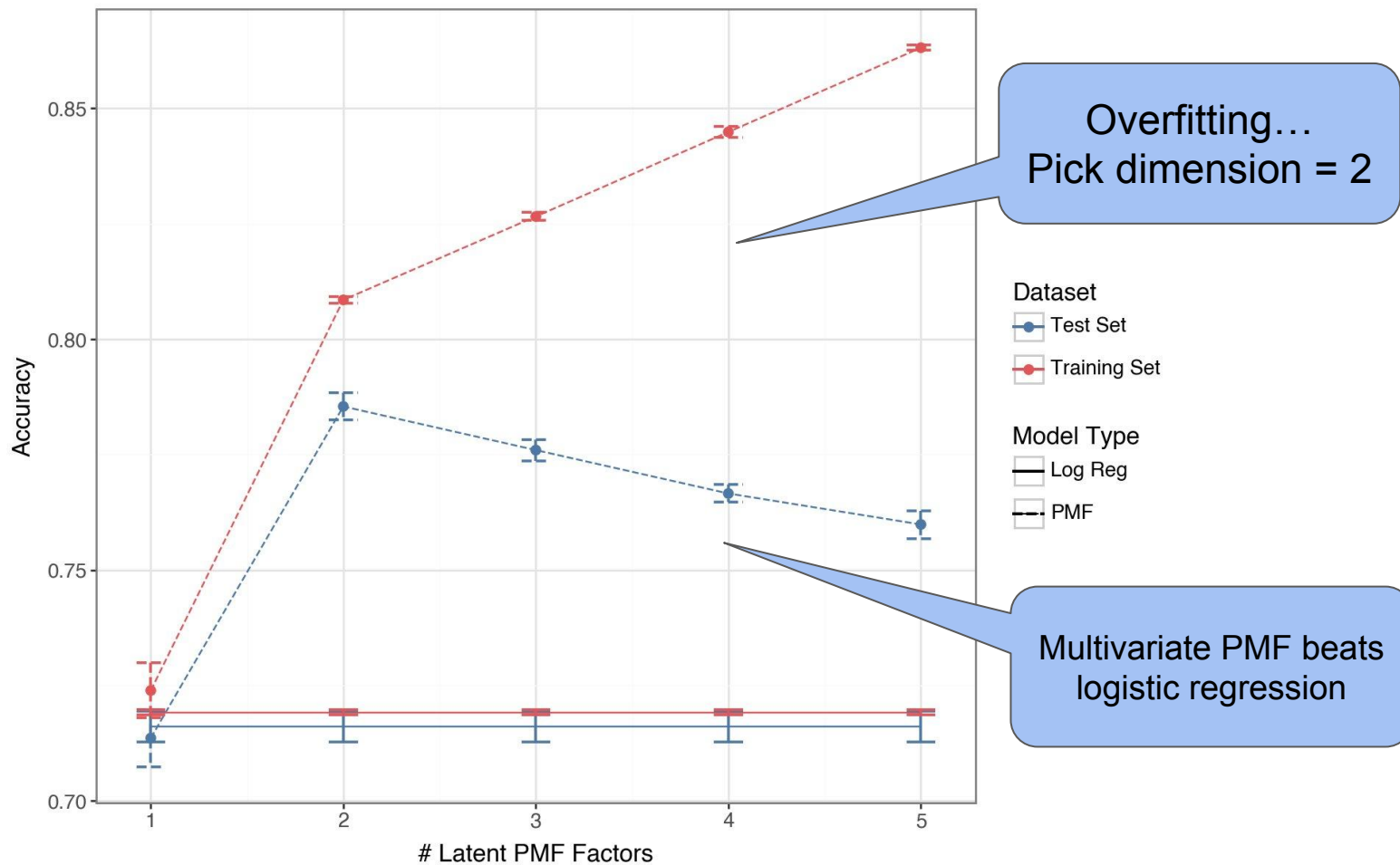
-1.29	-0.36	1.14	-0.44	0.79
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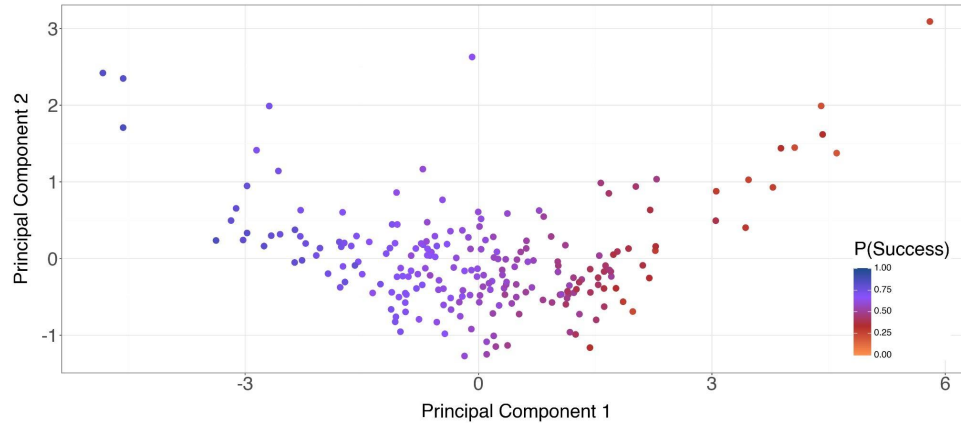
-4.6



Results



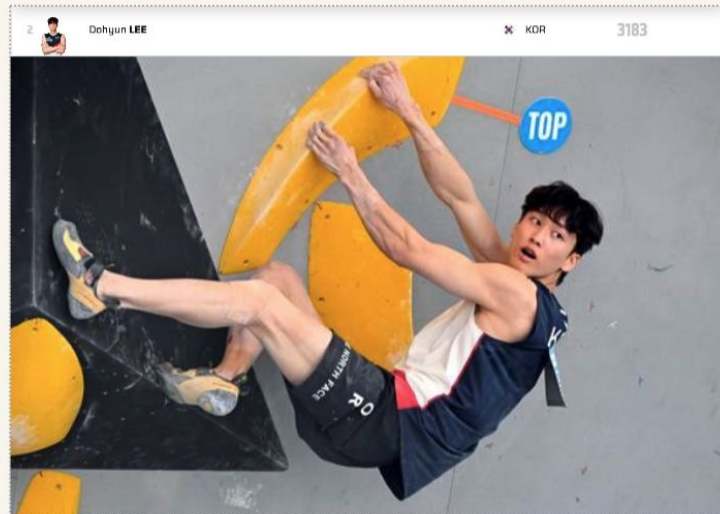
What have our embeddings learned?



PC1 = Overall strength
PC2 = Height? Power? Dyno?

	PC1			
	# Climbs	P(Success)	LR Coef	Height
PC1	-0.56	-0.94	-0.95	0.096
PC2	0.0043	-0.0012	0.027	0.33

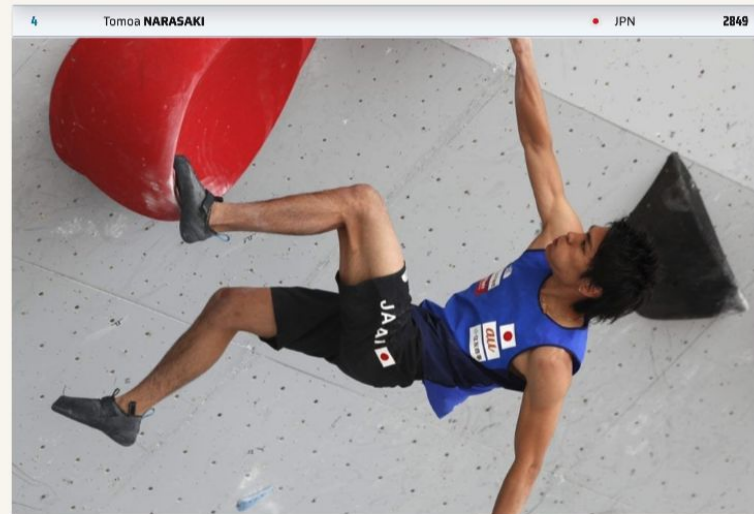
Example



Dohyun Lee

PC1: -2.75

PC2: 0.16



Tomoa Narasaki

PC1: -2.66

PC2: 1.99

What's next?

Use cases:

- Coaching: prioritize training
 - Routesetting: fair competitions
 - Forecasting: predict tournament winners
-
1. Interpretability, boulder problem features (CV?)
 2. Evolution of climber rating over time

Thanks!

arXiv

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[Submitted on 4 Nov 2024 (v1), last revised 8 Nov 2024 (this version, v2)]

Boulder2Vec: Modeling Climber Performances in Professional Bouldering Competitions

Ethan Baron, Victor Hau, Zeke Weng

Using data from professional bouldering competitions from 2008 to 2022, we train a logistic regression to predict climber results and measure climber skill. However, this approach is limited, as a single numeric coefficient per climber cannot adequately capture the intricacies of climbers' varying strengths and weaknesses in different boulder problems. For example, some climbers might prefer more static, technical routes while other climbers may specialize in powerful, dynamic problems.

To this end, we apply Probabilistic Matrix Factorization (PMF), a framework commonly used in recommender systems, to represent the unique characteristics of climbers and problems with

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baron22 · boulder2vec

Code Issues Pull requests Agents Actions Projects Security and quality Insights Settings

boulder2vec (main)

main · 3 Branches · 1 Tag

do to file Add file Code About

ZekeWeng added powerpoint

7220195 · last year

110 Commits

data

Created Climber Heights

2 years ago

models

added problem type to lr (#18)

2 years ago

models_lr

uploaded pmf 50, 100

2 years ago

results

Update replication instructions and delete old files (#26)

2 years ago

gignore

ignore bash script

2 years ago

Boulder2Vec.pdf

Add paper

2 years ago

Boulder2Vec.pptx

added powerpoint

last year

README.md

Update replication instructions and delete old files (#26)

2 years ago

climber_embeddings.py

Added Height to Embeddings

2 years ago

eda.py

EDA figures (#25)

2 years ago

eval.py

Update replication instructions and delete old files (#26)

2 years ago

figs.py

Update replication instructions and delete old files (#26)

2 years ago

Modeling climber performances in professional bouldering competitions

[bouldering](#) [bouldering analysis](#)
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Timeline

Activity

5 stars

2 watching

0 forks

Releases (1)

v1 (latest)

2 years ago

Packages

No packages published

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Contributors (3)

Reproducible Research Competition

Submit an abstract to present your sports analytics work in the Reproducible Research Competition at the annual Carnegie Mellon Sports Analytics Conference on Oct 23-24 2026!

In this year's competition, there are two tracks: (1) high school / undergraduate students and (2) graduate students / non-students. In addition to accepting submissions with innovative sports analytics methodology, we also accept data and software contributions that provide the public with easy access to sports data (e.g., a new R package to access data that was previously not readily available).

Timeline:

- **Friday August 14th 5:00 PM ET: Abstract submission deadline.**
- Monday August 17th: Selected abstracts are notified with directions for submitting papers.
- **Friday September 4th 5:00 PM ET: Paper submission deadline.**
- Friday October 2nd: Competition finalists are announced and invited to present at CMSAC on Oct 24th! Additional paper submissions will be invited to present work as posters.
- **Saturday Oct 24th: Finalists present at CMSAC and are awarded cash prizes!** Separate prizes for: (1) high school / undergraduate students and (2) graduate students / non-students.